

Point-to-Point Communication System

EK 210 Introduction to Engineering Design

Final Report

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Executive Summary

This project presents a portable, autonomous optical communication system that encodes and transmits messages point-to-point via a 650nm laser, requiring no WiFi, Bluetooth, or RF. Designed for classroom demonstrations of laser communication fundamentals, the system allows a user to enter a message via keypad, which is encoded and transmitted to a receiver unit that decodes the signal and displays it on an LCD screen. A 5V stepper motor automates the transmitter's directional aiming, and both units are housed in a laser-cut wood and 3D-printed enclosure powered by a portable battery. Performance testing demonstrated that increasing bit encoding time and laser drive current reduced the error rate to virtually zero under stable alignment conditions, achieving $\sim 98\%$ proper decoding rate at close range. The final system generally meets key design objectives, including eye-safe laser operation at 650nm, autonomous transmission with no human intervention, and a total bill of materials cost within a \$250 budget. Precise alignment between the transmitter and receiver remains a requirement for reliable operation, and indoor use is recommended to minimize interference from ambient light.

1 Introduction

Short-range wireless communication typically relies on WiFi, Bluetooth, or radio frequency, which are protocols that obscure the underlying physics of optical transmission. This project addresses that gap by designing a portable laser-based system for classroom demonstrations of point-to-point optical communication. Key design objectives are summarized in Appendix A: $\geq 95\%$ digit accuracy, eye-safe laser operation at $\sim 650\text{nm}$ (per IEC 60825-1),

transmission of a 10-digit message to a receiver at least 3 meters away in ≤ 30 seconds, compact footprint ($\leq 4,000 \text{ cm}^3$), and BOM cost within \$250. The following sections describe design decisions, the final prototype, and evaluation of results against these objectives.

2 Design

2.1 Important and Innovative Design Elements

The 650nm laser (X001FS32MZ) was selected for its narrow emission angle and eye-safe operation per IEC 60825-1. The received signal is amplified by an MCP6241 op-amp and digitized by an LM311P comparator for reliable bit detection. A key design iteration was increasing bit-encoding time and laser drive current, which reduced the error rate to virtually zero. Two additional 650nm lasers on the receiver side, paired with a 650nm bandpass filter, provide visible alignment confirmation and ambient light rejection, improving detection reliability. Directional aiming is automated via a 28BYJ-48 stepper motor and ULN2003 driver, which sweeps the transmitter until the receiver detects the beam.

2.2 Description of Final Prototype

The prototype comprises a transmitter and receiver, each in a laser-cut wood enclosure with 3D-printed optical mounts. The user enters a 10-digit message via keypad; the Arduino UNO encodes it as a binary pulse train and drives the 650nm laser while the stepper sweeps to find alignment. The transmitter is powered by a 9V battery regulated through two HW-131 breadboard power supply modules, which provide stable 5V rails for the Arduino UNO, stepper motor, lasers, keypad, and LCD. The BPW24R phototransistor detects the signal through the 650nm bandpass filter, amplified, digitized, and decoded by the Arduino Nano, then displayed on the TC1602A-01T LCD. The receiver runs off a 9V battery through the Arduino Nano’s onboard regulator. Schematics and drawings are in Appendices F and H.

3 Evaluation of Results

Testing was conducted over 10 trials; full data are in Appendix J. The system achieved 98% average decode accuracy, exceeding the $\geq 95\%$ target, with all trials completing under 30 seconds (avg. 24.2s). Autonomy was partially met: 8 of 10 trials required no human intervention (80%); the 2 exceptions involved minor alignment shifts. Eye-safe 650 nm operation was verified per IEC 60825-1, BOM cost (\$163.00) is within budget, and both enclosures fit

within the 4,000 cm³ limit. Battery life is estimated at ~ 1.5 hours under full load, sufficient for classroom demonstrations; see Appendix I for full calculations. Five of seven objectives were fully met; autonomy and long-range reliability remain areas for future work.

4 Lessons Learned

Early system integration proved more complex than anticipated, as components that worked in isolation introduced timing and signal conflicts when combined. A key technical finding was the laser signal's high sensitivity to minor alignment changes, which ultimately motivated the stepper motor integration. Several last-minute component changes reinforced the importance of locking in design decisions early, since late substitutions introduced new challenges. Fabrication at EPIC required planning ahead for machine availability, and clear ownership across hardware, firmware, and mechanical subsystems kept coordination manageable. Battery access and power management were overlooked in the current design; both units are designed such that the top panel of the housing can be removed to access and replace the batteries, though a more integrated solution should be considered in future iterations.

5 References

- Arduino documentation. <https://docs.arduino.cc/hardware/nano> — <https://docs.arduino.cc/hardware/uno-rev3>
- AI assistance: Claude (Anthropic) was used to assist in typesetting this document.
- BPW24R datasheet. <https://www.vishay.com/docs/81521/bpw24r.pdf>
- 28BYJ-48 datasheet. <https://www.mouser.com/datasheet/2/758/stepd-01-data-sheet-1143075.pdf>
- HW-131 power supply module. <https://www.handsontec.com/dataspecs/mb/YwRobot%20Power%20Supply.pdf>
- IEC 60825-1. https://shop.textalk.se/shop/ws26/40626/files/full_size_-_for_start_page_banner/iec60825-1%7Bed1.2%7Den.pdf
- LM311P datasheet. <https://www.ti.com/lit/ds/symlink/lm311.pdf>
- MCP6241 datasheet. <https://ww1.microchip.com/downloads/en/DeviceDoc/20001685E.pdf>
- SparkFun KEYPAD-3X4-12B-1. <https://www.sparkfun.com/products/retired/14662>
- TC1602A-01T datasheet. <https://cdn-shop.adafruit.com/datasheets/TC1602A-01T.pdf>
- ULN2003 datasheet. <https://www.ti.com/lit/ds/symlink/uln2003a.pdf>
- X001FS32MZ laser module. <https://www.adafruit.com/product/1054#technical-details>

A Objectives and Metrics

Objective	Description	Metric
Be accurate	Correctly decode the transmitted message on the receiver with high fidelity	$\geq 95\%$ digit decode accuracy at 50 cm
Be safe	Ensure all laser emissions remain within eye-safe limits at any distance	650 nm operation per IEC 60825-1
Be autonomous	Operate independently without external connections or user intervention during transmission	Zero human interventions per transmission
Be reliable	Consistently transmit and decode 10-digit messages with minimal errors	Transmit 10-digit message in ≤ 30 seconds with ≤ 2 errors
Be fast	Minimize total transmission time from input to display	Message delivered in ≤ 30 seconds
Be compact	Fit both transmitter and receiver units within portable enclosures	Total footprint $\leq 4,000 \text{ cm}^3$
Be cost-effective	Achieve all design goals using readily available components within budget	Total BOM cost $\leq \$250$

Table 1: Design objectives and metrics for the laser communication system.

B Glass Box Diagram

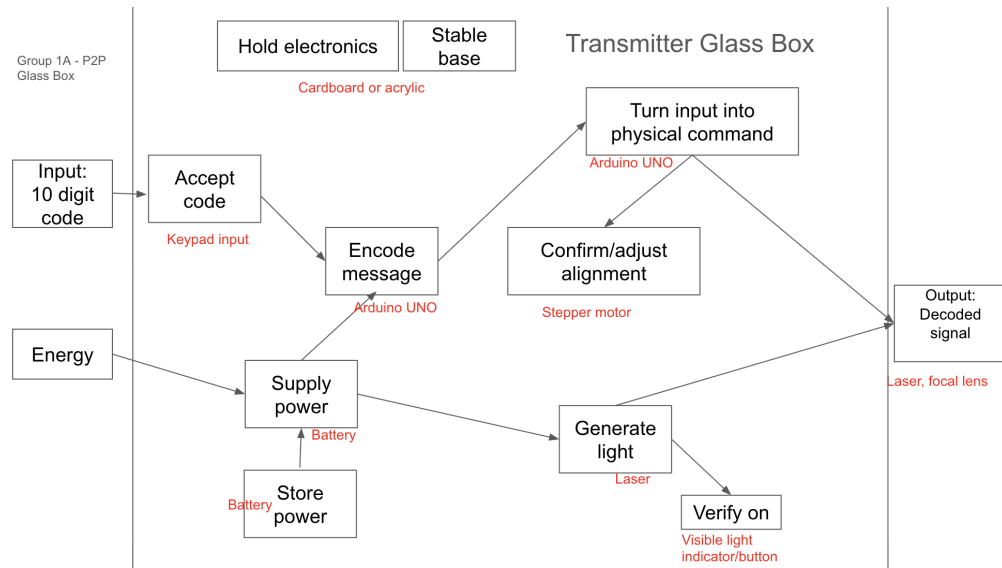


Figure 1: Transmitter Glass Box

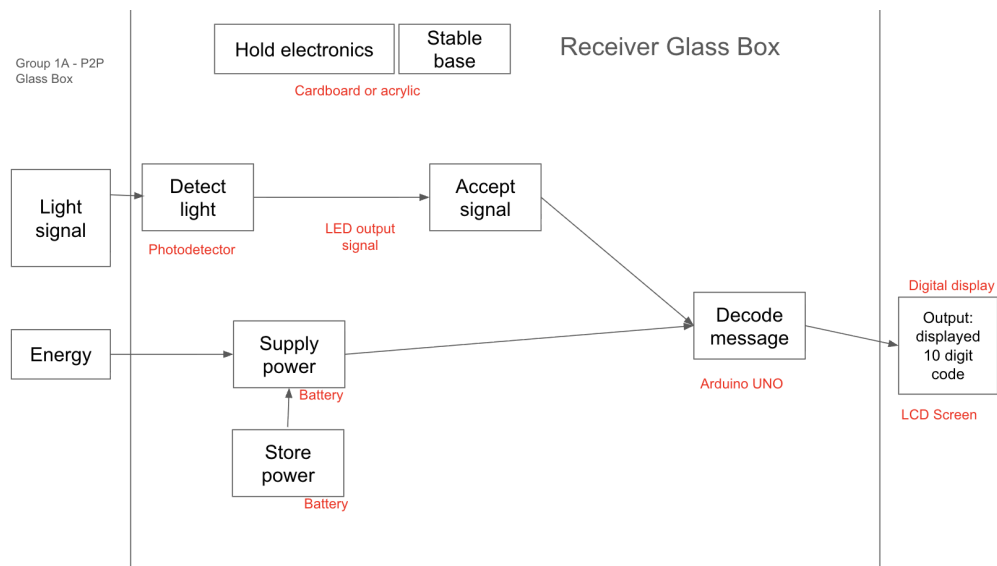


Figure 2: Receiver Glass Box

C PCC (Pairwise Comparison Chart)

Objective	Reliable	Safe	Compact	Accurate	Autonomous	Fast	Cost-eff.	Total	Rank
Be reliable	–	0	1	0	0	1	1	3	4
Be safe	1	–	1	0	1	1	1	5	2
Be compact	0	0	–	0	0	0	1	1	6
Be accurate	1	1	1	–	1	1	1	6	1
Be autonomous	1	0	1	0	–	1	1	4	3
Be fast	0	0	1	0	0	–	1	2	5
Be cost-effective	0	0	0	0	0	0	–	0	7

Table 2: Pairwise Comparison Chart with totals and final rankings.

D Morph Chart

Function	Option 1	Option 2	Option 3	Selected
Emit light	850nm IR LED	950nm IR LED	650nm laser	650nm laser
Detect signal	Photodiode	Phototransistor	IR receiver module	Phototransistor
Focus/collimate light	No lens	Collimating lens	Fresnel lens	No lens
Aim transmitter	DC motor	Servo motor	Stepper motor	Stepper motor
Process & control	Raspberry Pi	ESP32	Arduino	Arduino
User input	Push buttons	Touchscreen	Matrix keypad	Matrix keypad
Display output	Serial monitor	OLED display	LCD screen	LCD screen
Enclosure	3D printed PLA	Laser-cut acrylic	Laser-cut wood	Laser-cut wood
Power source	Powerbank	Rechargeable Li-Po	9V battery	9V battery

Table 3: Morph chart showing design alternatives considered for each system function.

E Bill of Materials

Part No.	Description	Qty	Unit Cost (\$)	Total (\$)
X001FS32MZ	650nm laser (1 TX, 2 RX)	3	5.00	15.00
BPW24R	Phototransistor (650nm)	1	3.00	3.00
—	650nm narrow bandpass filter	1	22.50	22.50
28BYJ-48	5V stepper motor	1	6.50	6.50
ULN2003	Stepper motor driver	1	1.50	1.50
HW-131	Breadboard power supply (TX)	2	2.00	4.00
Arduino UNO	Microcontroller (TX)	1	26.50	26.50
Arduino Nano	Microcontroller (RX)	1	22.50	22.50
MCP6241	Op-amp	1	1.50	1.50
LM311P	Voltage comparator	1	1.50	1.50
TC1602A-01T	LCD screen	2	12.50	25.00
KEYPAD-3X4-12B-1	Matrix keypad	1	5.00	5.00
—	9V alkaline battery	2	3.00	6.00
—	Laser-cut wood enclosure	2	7.50	15.00
—	3D printed mounts	2	3.50	7.00
			Total	163.00

Table 4: Final bill of materials.

F Electronics Sketch

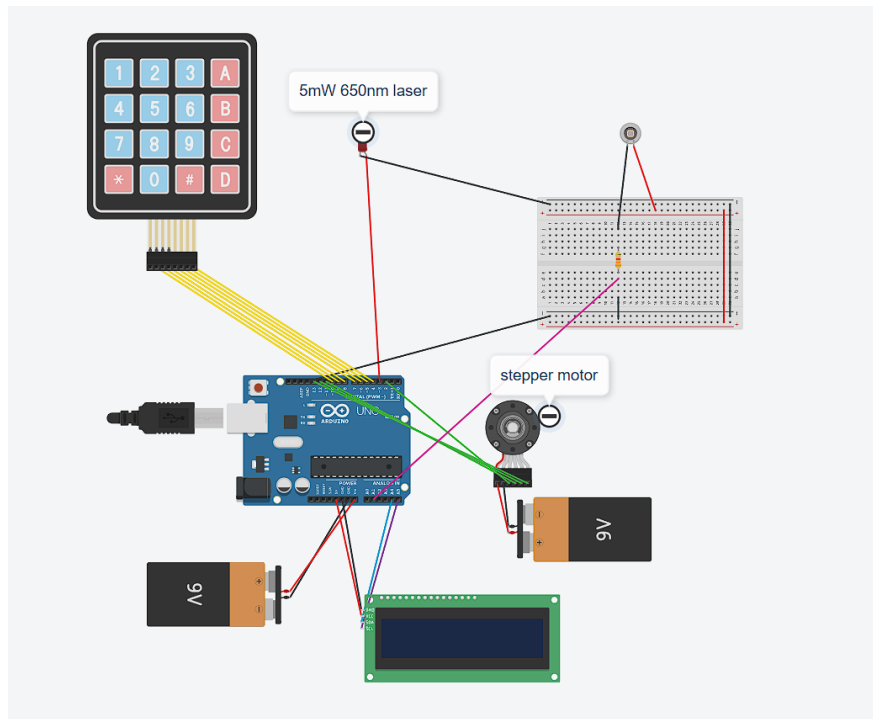


Figure 3: Electronics sketch of the transmitter

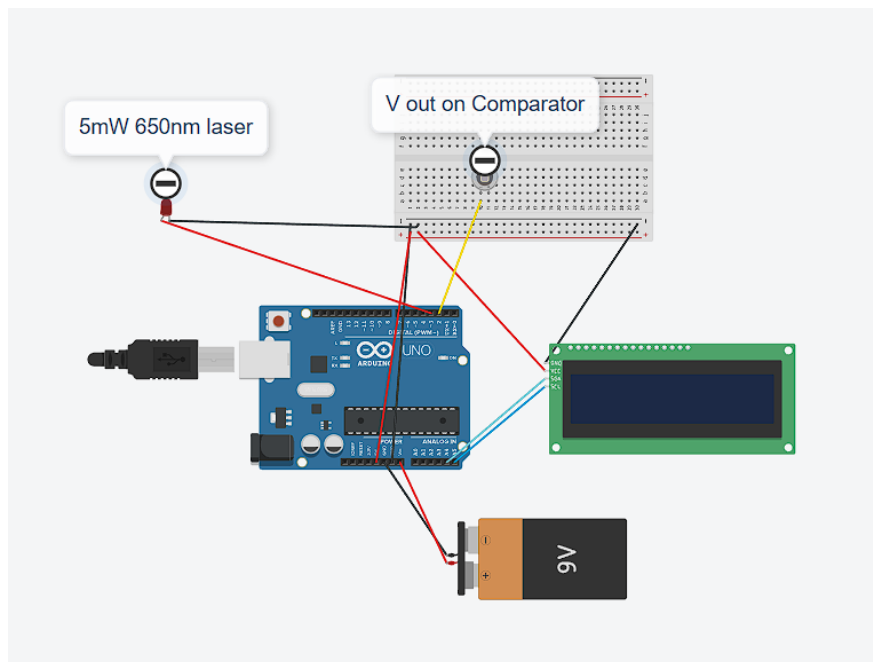


Figure 4: Electronics sketch of the receiver

G Code Flow Chart

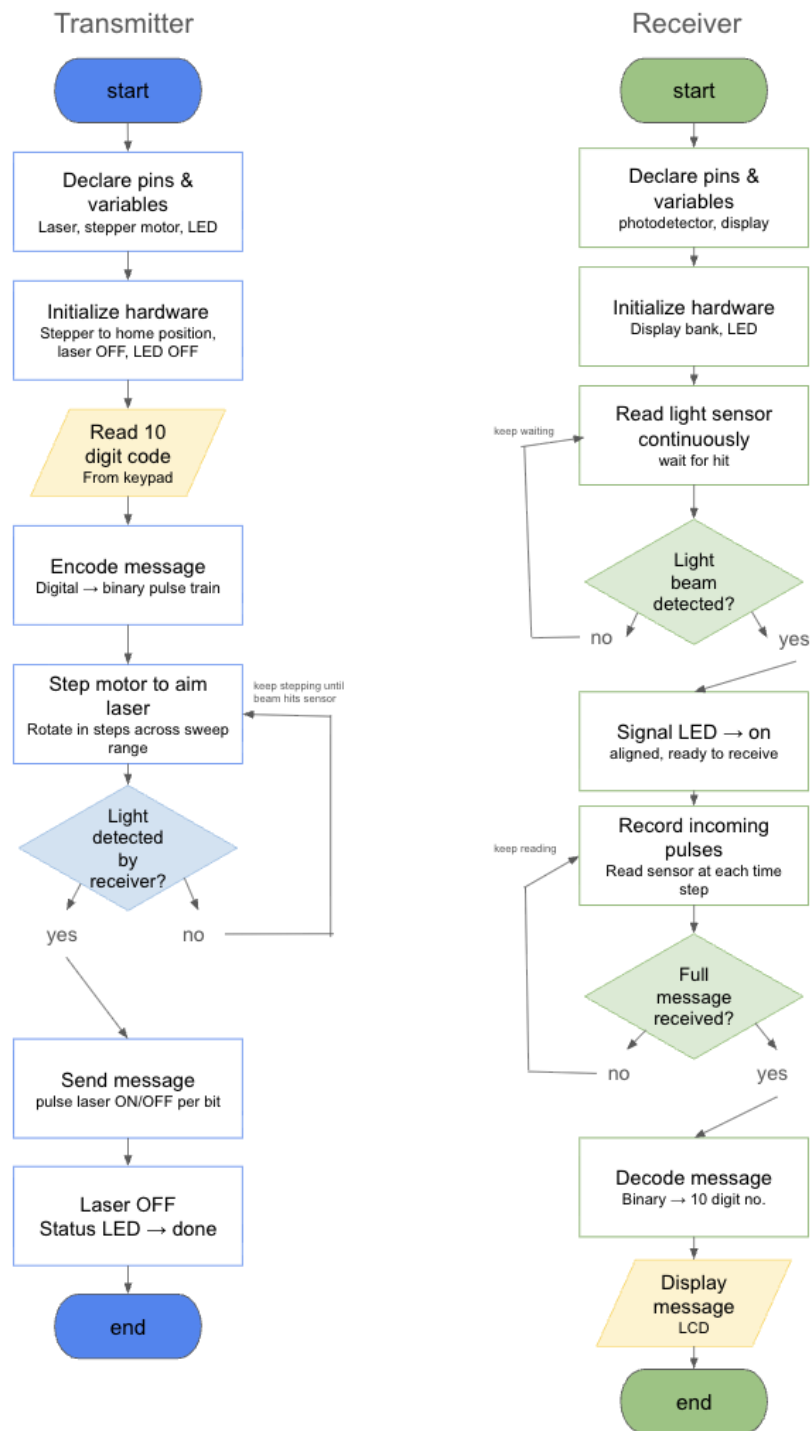


Figure 5: Code flow chart for transmitter and receiver.

H CAD or Hand Drawings

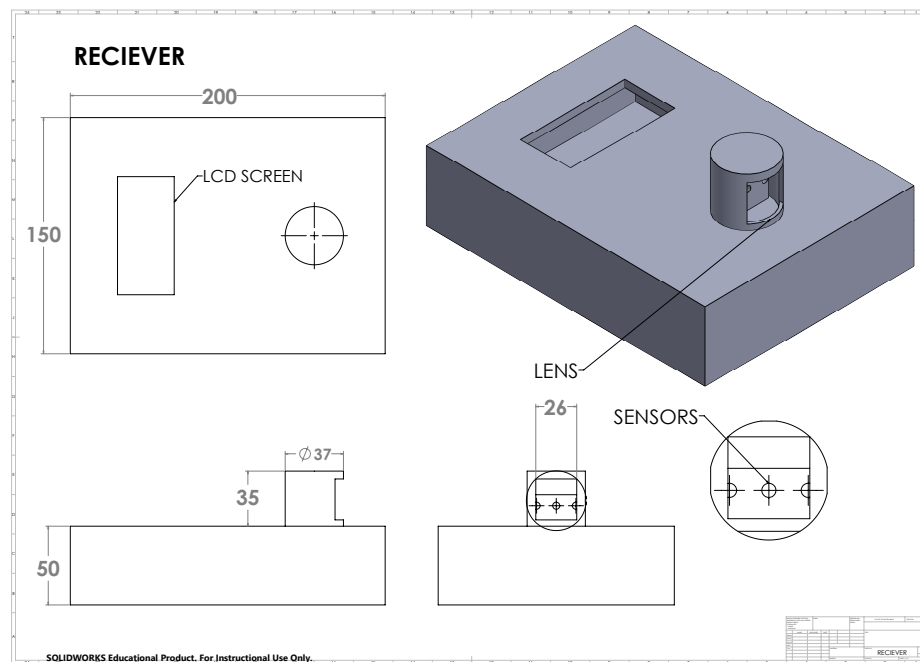


Figure 6: Final CAD design of the receiver.

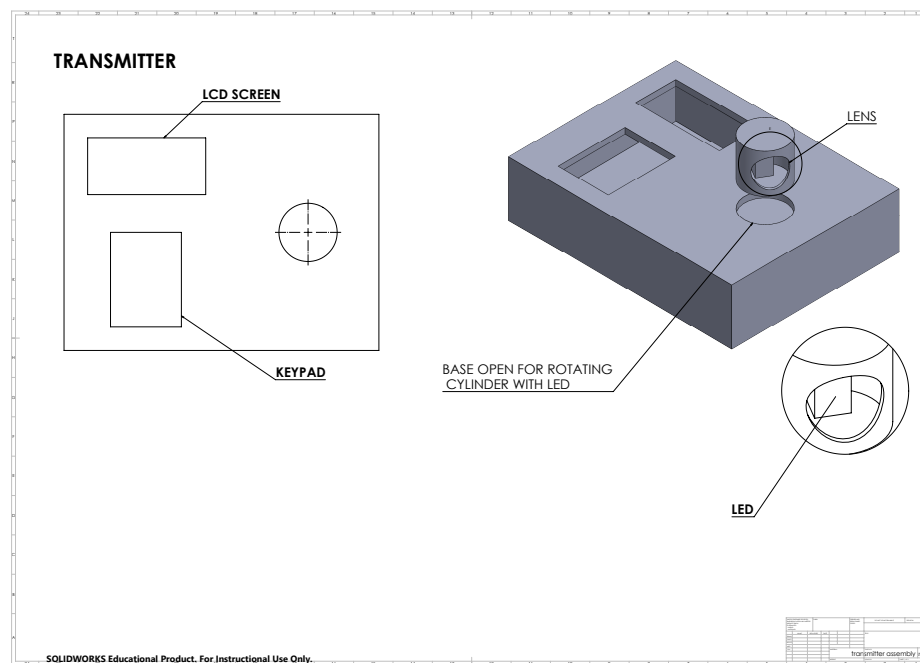


Figure 7: Final CAD design of the transmitter.

I Power Budget Calculations

All components are powered by 9V batteries. On the transmitter side, two HW-131 modules regulate the input to stable 5V rails; the receiver runs off a 9V battery through the Arduino Nano’s onboard regulator. Table 5 summarizes estimated current draw and power consumption per component.

Component	Voltage (V)	Current (mA)	Power (mW)
Arduino UNO	5	50	250
Arduino Nano	5	19	95
28BYJ-48	5	240	1200
ULN2003	5	7	35
HW-131 (x2)	9	10	90
X001FS32MZ (x3)	5	120	600
BPW24R	5	1	5
MCP6241	5	1	5
LM311P	5	6	30
TC1602A-01T (x2)	5	80	400
KEYPAD-3X4-12B-1	5	1	5
Total		535	2715

Table 5: Estimated power consumption per component.

The total estimated power draw is ~ 2.72 W. Accounting for a voltage regulator efficiency of $\sim 85\%$, the effective current draw from the 9V battery is:

$$I_{battery} = \frac{P_{total}}{V_{battery} \times \eta} = \frac{2715 \text{ mW}}{9 \text{ V} \times 0.85} \approx 355 \text{ mA} \quad (1)$$

A standard 9V alkaline battery has a nominal capacity of ~ 550 mAh, giving an estimated operating duration of:

$$t = \frac{C}{I_{battery}} = \frac{550 \text{ mAh}}{355 \text{ mA}} \approx 1.5 \text{ hours} \quad (2)$$

Note that the stepper motor dominates power consumption and is only active during the alignment phase. During transmission-only operation, current draw drops to approximately 186 mA, extending estimated battery life to ~ 3.0 hours.

J Plots and Tables — Testing Results

Accuracy

Trial	Digits Correctly Decoded	Accuracy (%)
1	10	100
2	10	100
3	9	90
4	10	100
5	10	100
6	10	100
7	9	90
8	10	100
9	10	100
10	10	100
Average	9.8	98%

Table 6: Digit decode accuracy over 10 transmission trials. Each trial transmits a 10-digit message; accuracy is calculated as the percentage of digits correctly decoded.

Accuracy by Digits Sent

Digits Sent	Avg Digits Correctly Decoded	Avg Accuracy (%)
4	4.00	100
5	5.00	100
6	5.94	99
7	6.86	98
8	7.76	97
9	8.64	96
10	9.60	96
Avg of Averages	6.83	98

Table 7: Average digit decode accuracy as a function of message length. Results show a slight decrease in accuracy with increasing digits sent; all values remain above the $\geq 95\%$ target.

Autonomy

Trial	Human Interventions	Notes
1	0	
2	0	
3	1	Alignment shifted mid-sweep
4	0	
5	0	
6	0	
7	0	
8	1	Manual realignment required
9	0	
10	0	
Total	2	

Table 8: Number of human interventions required per transmission trial. A value of zero indicates fully autonomous operation.

Speed

Trial	Transmission Time (s)
1	24.3
2	22.8
3	27.1
4	23.5
5	21.9
6	24.7
7	25.3
8	26.8
9	22.4
10	23.6
Average	24.2

Table 9: Total transmission time per trial, measured from keypad entry to message display on the receiver LCD. Target: ≤ 30 seconds.

Transmission Time by Digits Sent

Digits Sent	Avg Transmission Time (s)
4	12.5
5	14.8
6	17.2
7	19.6
8	22.1
9	24.4
10	26.8
Avg of Averages	19.6

Table 10: Average transmission time as a function of message length. All values remain well below the ≤ 30 second target.